

```

clc
clear all
syms x y ax ay real
f = input('Enter the function f(x,y):');
fx = diff(f,x); fy = diff(f,y);
[ax, ay] = solve(fx,fy);
ax=double(ax); ay=double(ay); r=length(ax);
fxx = diff(fx,x);
fxy=diff(fx,y);
fyy=diff(fy,y);
D = fxx*fyy - fxy^2;

a1=max(ax); a2=min(ax); ex=0.5;
b1=max(ay); b2=min(ay); ey=0.5;

s=fsurf(f,[a2-ex,a1+ex,b2-ey,b1+ey],'g','EdgeColor','none' );

s.FaceAlpha = .65;
box on
hold on
for r1=1:r

    T1=subs(subs(D,x,ax(r1)),y,ay(r1)); T1=double(T1);
    T2=subs(subs(fxx,x,ax(r1)),y,ay(r1)); T2=double(T2);
    T3=subs(subs(f,x,ax(r1)),y,ay(r1)); T3=double(T3);

    if (T1 == 0)
        fprintf('The point (%d,%d) needs further
investigation. \n', ax(r1),ay(r1))
    elseif (T1 < 0)
        fprintf('The point (%d,%d) is a saddle point. \n', ax(r1),ay(r1))
        plot3(ax(r1),ay(r1),T3,'b.','markersize',30);

    else
        if (T2 < 0)
            fprintf('(%d, %d) is a point of maxima. \n', ax(r1),ay(r1))
            fprintf('The value of the function at the point of maxima is %d. \n',
T3)
            plot3(ax(r1),ay(r1),T3,'r.','markersize',30);
        else
            fprintf('(%d, %d) is a point of minima. \n', ax(r1),ay(r1))
            fprintf('The value of the function at the point of minima is %d. \n',
T3)
            plot3(ax(r1),ay(r1),T3,'k.','markersize',30);
        end
    end
end
End

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clc
clear all
syms x y lam real
f = input('Enter f(x,y) to be extremized : ');
g = input('Enter the constraint function g(x,y):');

F = f+ lam*g
Fd = jacobian(F,[x y lam])

[ax, ay, alam] = solve(Fd, x, y, lam);
ax = double(ax);
ay = double(ay);

T = subs(f,{x,y},{ax,ay});
T = double(T);
epxl = min(ax);
epxr = max(ax);
epyl = min(ay);
epyr = max(ay);
D = [epxl-1.5 epxr+1.5 epyl-1.5 epyr+1.5]
fcontour(f, D, 'LevelList', -12:1:12)
axis equal
hold on
h = fimplicit(g);
set(h,'Color',[1,0.7,0.9])

for i = 1:length(T)
    fprintf('The function f(x,y) takes on its extreme value on the
g(x,y) at (%1.3f,%1.3f).', ax(i), ay(i))
    fprintf('The value of the function is %1.3f\n', T(i))
    plot3(ax(i), ay(i), T(i), 'k.', 'markersize', 15)
end

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clc
clear
syms t x y z
F=input('Enter the f vector as i and j order in vector form:');
rbar = input('Enter the r vector as i and j order in vector form:');
lim=input('Enter the limit of integration:');
vecfi=input('Enter the vector field range'); % depends on problem
drbar=diff(rbar,t);
sub = subs(F,[x,y,z],rbar);
f1=dot(sub,drbar)
int(f1,t,lim(1),lim(2))
P = inline(vectorize(F(1)), 'x', 'y', 'z');
Q = inline(vectorize(F(2)), 'x', 'y', 'z');
R = inline(vectorize(F(3)), 'x', 'y', 'z');
x = linspace(vecfi(1),vecfi(2), 7); y = x;
z=x;
[X,Y,Z] = meshgrid(x,y,z);
U = P(X,Y,Z);
V = Q(X,Y,Z);
W = R(X,Y,Z);
quiver3(X,Y,Z,U,V,W,1.5)
hold on
fplot3(rbar(1),rbar(2), rbar(3),[lim(1),lim(2)])
axis on

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```

clc

clear all

syms x t k X(x) T(t) C1

u=X(x)*T(t);

ux=diff(u,x); ut=diff(u,t);

pde=3*ux+2*ut;

pde1=expand(pde/u);

deX=subs(pde1==k,diff(T(t), t),0);

cpde1=subs(pde1,{diff(X(x), x),diff(T(t), t)},{0,0});

deT=subs(pde1-cpde1==-k,diff(X(x), x),0);

X(x)=dsolve(deX); X(x)=subs(X(x),'C1',1);

T(t)=dsolve(deT);

ic1=X(x)==exp(-x);

ic2=T(0)==4;

[k,C1]=solve(ic1,ic2,[k,C1]);

u=subs(X(x)*T(t));

fprintf('\nSolution by separation of variables method is\n')

fprintf('\nu(x,t) = %s\n',u)

fsurf(u,[0 1 0 1])

xlabel('x');ylabel('y');zlabel('u');

% Verification

pde=3*ux+2*ut;

uxs=diff(u,x)

uts=diff(u,t)

v=subs(pde,[ux ut],[uxs uts])

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